Machine Learning-Driven Greenhouse Management: A Comparative Analysis of Prediction Models

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***Abstract*—Modern agriculture increasingly relies on smart greenhouses to create controlled environments that optimize crop growth. Effective management of these systems requires accurate prediction of key environmental parameters, robust decision-making processes, and a strong focus on enhancing operational efficiency and sustainability. This study presents a comprehensive evaluation of two machine learning models, the Random Forest Regressor (RFR) and the Gradient Boosting Regressor (GBR), to predict greenhouse conditions for tomato cultivation and manage critical devices such as heating, lighting and irrigation. The models were assessed based on root mean squared error (RMSE), R-squared (R^2), and other relevant performance metrics.These results demonstrate the robustness and generalization capacity of both models. RFR exhibited slightly better predictive performance for heating and irrigation, while both models achieved near-identical accuracy for lighting. The high precision of both algorithms highlights their potential for real-time greenhouse management.**

***Index Terms*—Smart greenhouse, machine learning, precision agriculture, sustainability.**

I. Introduction

The growing demand for sustainable food security, preserving the environment, natural ecosystems, and landscapes has created an increased need for integrated, scalable agricultural solutions [1]. The evolution of agriculture from traditional methods to precision agriculture, often referred to as Agriculture 4.0, underscores the pivotal role of advanced technologies and smart greenhouses in optimizing various aspects of modern farming. Smart greenhouses are enclosed structures designed to provide optimal conditions for crop cultivation while protecting plants from external threats [2], [3]. These systems integrate Internet of Things (IoT) and machine learning (ML) technologies to enhance environmental management and resource efficiency, paving the way for sustainable agricultural practices. By enabling real-time decision-making, IoT reduces supply chain losses [4] and provides cost-effective solutions through predictive capabilities that anticipate future resource requirements [3]. Due to the dynamic and nonlinear nature of greenhouse environments [2], accurate prediction of internal parameters is needed to evaluate environmental-control strategies for crop growth [5]. Emerging technologies such as wireless sensor networks (WSN), Internet of Things (IoT), artificial intelligence (AI), deep learning (DL), and robotics are the main ongoing technologies achieving automation and real-time decision-making in agriculture [6]. ML has shown significant potential in smart farming to perform agricultural tasks [7]. By analyzing historical data, ML techniques enable the creation of fully autonomous greenhouses. Applications of ML in agriculture include prediction of yield [8], disease detection [9], weed detection [10], crop quality [11], species livestock management [12], and water and soil management [13] among authors. However, the adoption of ML in smart greenhouses remains in its early stages, with limited com-

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parative analyses evaluating the performance of ML models in such environments [14]. This paper addresses this gap by comparing the effectiveness of two ML algorithms, RFR and GBR, for predicting key greenhouse parameters and managing critical devices. Tomato, the key focus of this study, is highly sensitive to environmental conditions, which are crucial to maximizing yield and meeting market requirements [15] . Temperature is the most critical factor in tomato production, as the crop is highly sensitive to thermal variations. Low temperatures (below 10°C) hinder growth, cause difficulties in fruit setting, and lead to fasciated flowers. Conversely, temperatures exceeding 30°C prevent the synthesis of lycopene, the pigment responsible for the red color of tomatoes [16]. Additionally, maintaining a relative humidity of 75% is optimal to ensure fruits of good size while preventing diseases [16]. This paper focuses on the development and evaluation of predictive models for greenhouse devices. By employing RFR and GBR, we aim to establish a robust decision-support system that ensures efficient and sustainable greenhouse management. These studies reinforce the significance of ML in sustainable agriculture and pave the way for precise and more responsive agriculture.

II. Related work

Recent studies in IoT and ML have significantly influenced smart greenhouse monitoring. Machine learning techniques are increasingly used to study complex environments with high temporal and spatial variability [17], gaining significant attention for their ability to predict and manage greenhouse microclimate parameters. Recent studies have explored various models for predicting environmental parameters. One notable work is the study by [18], which developed a system using artificial neural networks (ANN) to create data-driven prediction models of the greenhouse microclimate. This system was experimentally validated by monitoring temperature and humidity in a tomato crop greenhouse in Mexico over six months. Similarly, [19] combined Fuzzy Neural Networks (FNN) and ANN to control temperature and humidity variations. The system was designed to regulate various devices within the greenhouse, including wet curtain fans, shade nets, circulating fans, sunroofs, heaters, and micro-mist humidifiers. These devices play a crucial role in maintaining optimal growing conditions for plants. The results demonstrated the system's stability, responsiveness, and reliability in real-world greenhouse conditions. Another study by [4] integrated IoT and Convolutional Neural Networks (CNNs) to analyze physical variables and detect plant diseases, achieving an average accuracy of 95%. Additionally, [20] proposed a framework combining weather forecast data, numerical models (HYDRUS-1D), and ML models like RF to predict soil temperature and water content. The RF model provided the most accurate predictions, optimizing water management in dynamic conditions. Other notable contributions include [21] , which used image processing and computer vision to evaluate plant leaf disease using CNNs, and [22] , which conducted a comparative analysis of Bi-LSTM, ANN, Gradient Boosting Machine (GBM), and RF models for forecasting temperature, humidity, and CO2 levels. Ensemble methods like GBM and RF achieved superior performance, highlighting the challenges of adapting ML models to dynamic greenhouse environments. The article by [22] discusses the implementation of an Intelligent Data Analytics Framework for Precision Farming using IoT and machine learning algorithms. Each layer plays a critical role in collecting, processing, and visualizing data. The fog layer focuses on data analytics and decision-making. Machine learning algorithms such as Support Vector Machine (SVM) and Multi-layer Perceptron Neural Network (MLP) are employed for regression modeling to predict outcomes related to pump on/off times, ventilation fan duration, and light levels. The experimental model conducted tests with Gerbera and Broccoli crops under various conditions. [17] conducted a comparative analysis of classical, shallow, and deep learning ML models to predict greenhouse gas (GHG) emissions. The study found that the LSTM model achieved the highest predictive accuracy (R^2=0.87,R^2 = 0.87,R^2=0.87) compared to other ML models such as RF and SVM. [22] explored the potential of ML models, particularly RF, for predicting N2O fluxes in intensive agricultural systems. By combining ML with biophysical models, the authors demonstrated a significant improvement in N2O emissions predictions compared to traditional models. Moreover, [23] proposed an AI-driven pest prediction system for managing Parlatoria Date Scale (PDS) in date palms. A stacking-based weighted ensemble model was developed and achieved high predictive accuracy for managing pest populations. Despite the extensive use of ML for various applications, most studies focus on individual model performance without direct comparisons under real greenhouse conditions. This work distinguishes itself by systematically comparing the performance of RFR and GBR in predicting greenhouse parameters and controlling devices. This study provides an empirical evaluation of the models' effectiveness in a real greenhouse setting.

III. Methodology

Effective management of greenhouse conditions relies on accurate prediction of environmental variables to ensure optimal resource utilization. This paper proposes a predictive control system that employs RFR and GBR to predict key parameters. The predicted values are then used to regulate greenhouse devices such as heating, lighting, and irrigation. The methodology consists of the following steps (Figure 1):

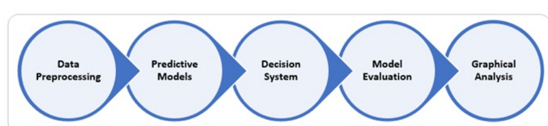

Fig. 1: Methodology design.

A. Data Preprocessing:

The data used in this study were collected from a commercial greenhouse cultivated with tomato crops, located in the Agadir region near the Atlantic Moroccan coast. This greenhouse, with a total area of 338 m^2 and a density of 1.8 plants per m^2, is a single-chapel structure covered with polyethylene plastic. The dataset corresponds to the month of March and includes measurements taken every 15 minutes. Key parameters include :

- Internal and external air temperatures (Tin, Tout)
- Internal and external relative humidities (Rhin, Rhout)
- Outside wind speed (WS) and direction (WD)
- Global solar radiation (G)

The data collected were normalized to ensure compatibility with predictive algorithms and improve the accuracy of the model.

B. Predictive Models:

Two models, RFR and GBR, were employed to predict future environmental conditions. These predictions were used to proactively control the greenhouse environment.

a) Random Forest Regressor:
RF was proposed by as [24] a combination of tree predictors; each tree depends on the values of a random vector sampled independently and with the same distribution for all trees in the forest [24]. It builds a “forest” by generating several trees from subsamples of the dataset, then combine their average predictions to produce a robust result. This combination reduces error in classification and regression tasks thanks to the use of bootstrap aggregation, or bagging [25] . RFR excels in reducing overlearning thanks to its ensemble strategy while maintaining high accuracy even with noisy data [26] . The diagram (Figure 2) illustrates how an RFR makes predictions. The observed data points passed through multiple decisions. Each decision tree (Tree^2, Tree^2, ..., Tree^2) independently processes the data.

- **Brown Nodes (Branches):** Decision points where data is split based on specific features (e.g., temperature, humidity).
- **Green Nodes (Leaves):** Terminal nodes that store the final prediction for a given path in the tree.

Each tree’s prediction is combined by averaging all outputs to produce a final prediction. The standard deviation of these predictions estimates the uncertainty. The main benefit of using RF is that it requires less training time and has high accuracy. [27] .

b) Gradient Boosting regressor:
The family of boosting methods is based on a novel, constructive strategy of ensemble formation, where the primary notion of boosting is to add new models to the ensemble sequentially, while the popular ensemble techniques, such as random forests, rely on simple averaging of models in the ensemble [29] .

GBR builds trees sequentially. Each tree learns from the errors of its predecessor, thus improving overall prediction accuracy [30]. Unlike RF algorithms, which use independent trees, the learning procedure consecutively fits new models to provide a more accurate estimate of the response variable.

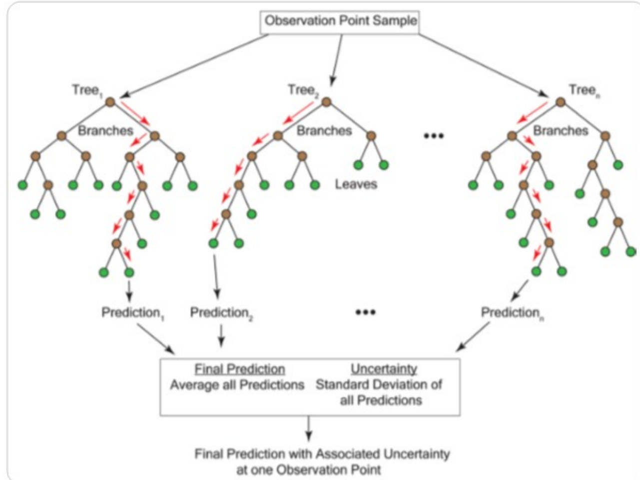

Fig. 2: Prediction steps using RFR from [28] .

As shown in Figure 3, the process starts with a data set that contains input samples X (features) and their corresponding target values Y. A decision tree is trained on the initial dataset to generate the first prediction. The residual (R1) is computed as the difference between the true values and the predicted values. This process is repeated iteratively, with each new weak model trained on the residuals (Rn-1) from the previous step. After N iterations, the outputs of all weak models are combined to produce the final prediction (Y). This iterative approach allows the GBR to be concise and appropriate for a scientific context [31].

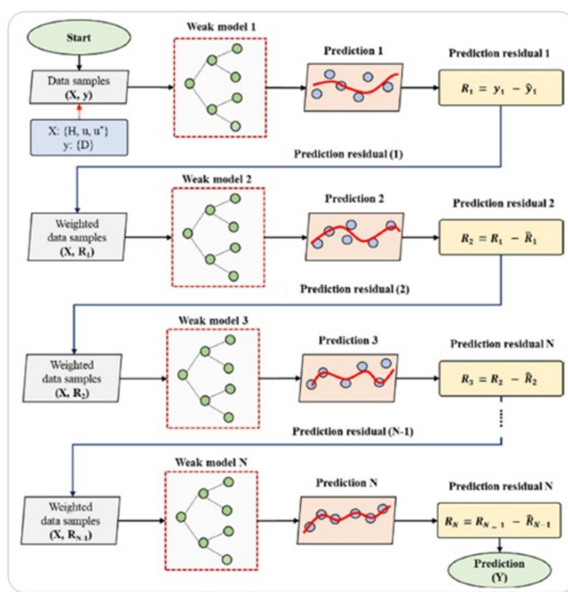

Fig. 3: Prediction steps using GBR from [30]

c) Why use the two algorithms?:
Based on [27] and [31] studies, this table (Table 1) highlights the complementary features of RFR and GBR, showcasing their robustness, accuracy, and applicability. As shown, RFR

TABLE I: Comparative table of RFR and GBR

Aspect	***Random Forest Regressor (RFR)***	***Gradient Boosting Regressor (GBR)***	***Complementarity***
Robustness to noise	+++	+	RFR stabilizes predictions in the presence of noise.
Prediction accuracy	++	+++	GBR refines predictions where RFR provides general reliability.
Training speed	+++	+	RFR offers quick results, while GBR further improves them.
Handling missing data	+++	+	RFR is more tolerant of raw, incomplete data.
Capturing complex relationships	++	+++	GBR excels in modeling subtle, complex patterns.
Resistance to overfitting	+++	++	RFR reduces overfitting risks effectively.

provides robustness, speed, and resistance to overfitting, while GBR excels in precision and modeling complex relationships. Together, they offer a balanced and effective solution for greenhouse prediction. The combination of RFR and GBR is complementary because RFR provides a reliable baseline by reducing overfitting risks, while GBR fine-tunes the model by iteratively minimizing residual errors. Together, they ensure a balance of robustness and precision, making them ideal for accurate temperature prediction in dynamic and complex environments such as greenhouses.

C. Decision System:

The decision-making process was tailored to the specific needs of tomato crops, which require: temperature (18-27 ° C), humidity (60- 70%) and light (12-16 h/day) [32], [33]. Artificial lighting supplements natural light when solar radiation drops below 400 W/m² [34]. Based on these thresholds, a set of control rules was established:

D. Model Evaluation:

The system evaluates model performance using error metrics such as R-squared (R^2) and the root mean square error (RMSE). Given by:

$$R^2 = 1 - \frac{\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{n}(y_i - \bar{y})^2} \tag{1}$$

$$RMSE = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2} \tag{2}$$

where $\hat{y}_i$ presents the predicted value, y_i the actual value, and n the number of observations.

The RMSE and R^2 are widely used metrics to evaluate model performance [34] [35]. While RMSE depends on the sum of squared errors and is sensitive to error variability, evaluates the proportion of variance explained by the model and considered more truthful in evaluating regression analysis [36].

E. Graphical Analysis:

To validate the predictive control system, simulations were developed in Python. Graphs comparing real and predicted values for heating, lighting, and irrigation were generated to assess model reliability and accuracy. These visualizations confirmed the models' effectiveness for real-time application in greenhouse management.

IV. Results and Analysis

The evaluation results for the RFR and GBR models for each process across the key greenhouse processes are presented below. The models' performance was assessed using RMSE and R-squared values. Table 2 summarizes the results: Heating: RFR displayed superior precision in predicting temperature variations, crucial for maintaining optimal growing conditions for tomatoes. With a lower RMSE (0.0302) and higher R^2 (0.978), RFR outperformed GBR (RMSE: 0.0363, R^2: 0.968) in modeling greenhouse temperature dynamics for tomatoes production.

TABLE II: Performance Comparison of Models for Different Processes

Process	***Model***	***RMSE***	***R²***
Heating	RFR	0.0302	0.9782
	GBR	0.0363	0.9685
Lighting	RFR	0.0007	0.9999
	GBR	0.0008	0.9999
Irrigation	RFR	0.0387	0.9843
	GBR	0.0473	0.9764

Lighting: Both models achieved nearly identical results, with exceptionally high R^2 values (0.9999), underscoring their effectiveness in handling lighting requirements. The negligible difference between RFR and GBR indicates that either model can be reliably used for light intensity prediction.

Irrigation: RFR also excelled in predicting humidity variations, with a slightly lower RMSE (0.0387) and higher R^2 (0.984) compared to GBR (RMSE: 0.0473, R^2: 0.976). This suggests that RFR is marginally better at modeling greenhouse humidity fluctuations, making it a strong candidate for precise irrigation control.

These results demonstrate the robustness and generalization capacity of both models, with RFR consistently achieving slightly better predictive performance in heating and irrigation. The high R^2 scores for the lighting process confirm the model's ability to capture sensor data variability accurately, validating the effectiveness of both RFR and GBR in automated greenhouse management. RFR proved robust in handling noisy data and achieving general reliability, while GBR remained competitive in modeling complex patterns. By reducing prediction errors, both models contribute to optimizing resource usage, minimizing energy consumption, and improving crop productivity in smart greenhouses. Based on these results, it is crucial to select the most suitable model for each specific function, such as heating, lighting, and irrigation, taking into account the unique characteristics and requirements of each process in smart greenhouses. This selection ensures optimal performance by aligning the strengths of the model with the specific needs of the tasks. Figures 4 and 5 illustrate the comparison between the real and predicted values for heating, lighting, and irrigation. As shown in the figures, the predicted values for heating, lighting, and irrigation match closely with the actual values. These graphs validate the reliability of the models and their suitability for real-time application in greenhouse management.

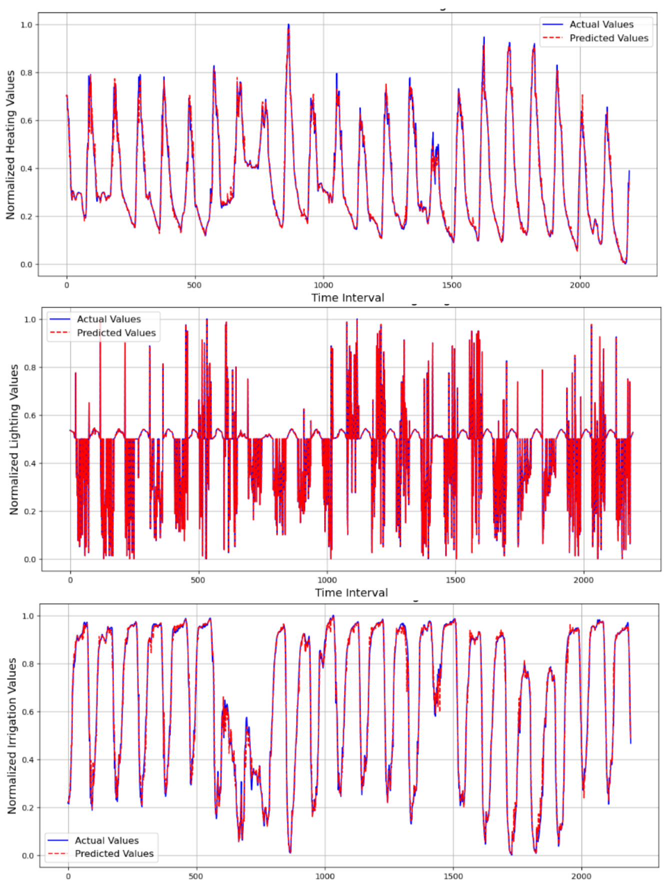

Fig. 4: Model performance of RFR algorithm.

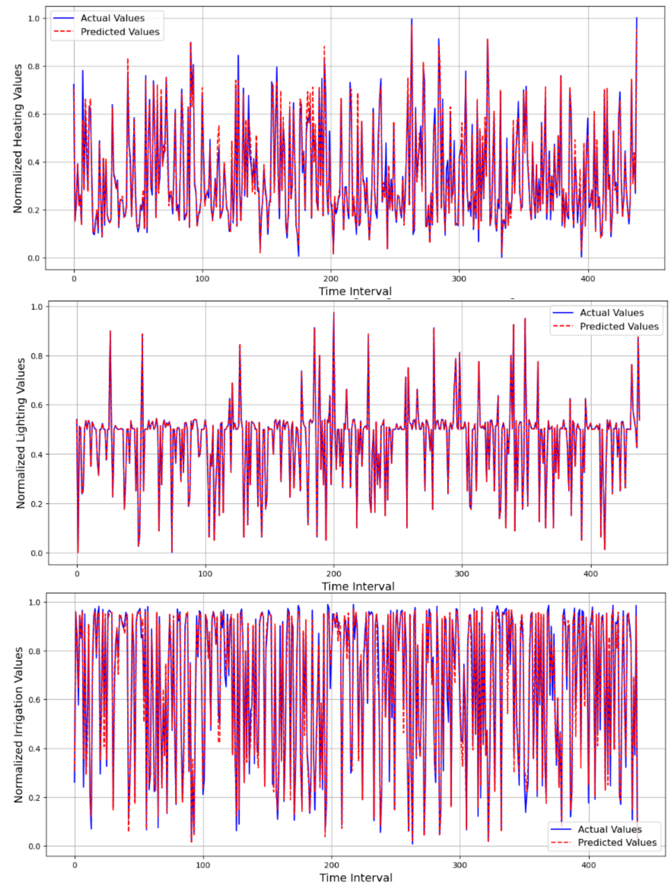

Fig. 5: Model performance of GBR algorithm.

In addition to the process-specific evaluation, a broader comparison was made between the overall performance of the Random Forest (RFR) and Gradient Boosting (GBR) models. This global comparison incorporated additional metrics, including MAE (mean absolute error), MSE (mean squared error), and R2, to provide a more comprehensive analysis of the models' performance across all greenhouse processes.

TABLE III: Global Model Comparison

Model	MAE	MSE	R^2
Random Forest	0.0187	0.0009	0.9782
Gradient Boosting	0.0237	0.0013	0.9685

The global comparison reveals that Random Forest (RFR) slightly outperforms Gradient Boosting (GBR) across all metrics. Specifically, RFR exhibits lower MAE and MSE values, suggesting that it generates fewer prediction errors on average. Furthermore, the higher R2 value (0.9782) indicates that it explains more of the variance in the data compared to GBR (R^2: 0.9685), highlighting its superior fit for the overall greenhouse data.

Despite these differences, both models performed admirably in predicting key greenhouse parameters, and their complementary strengths make them valuable tools for smart greenhouse applications. However, further research is needed to explore hybrid models, which integrate machine learning (such as RFR and GBR) with biophysical models based on physical and biological principles, and could improve prediction accuracy by considering both experimental data and natural mechanisms. This would optimize resource usage and improve the management of smart greenhouses. Machine learning models can be used to predict immediate outcomes based on sensor data, while biophysical models can simulate the physical and biological behaviors that influence those outcomes over time. This dual approach can lead to more accurate predictions that align with real-world processes.

V. Conclusion and Future Works:

This research highlights the transformative potential of machine learning in precision agriculture, promoting resource efficiency and sustainability. The complementary strengths of RFR and GBR make them valuable tools for smart greenhouse applications. RFR provides robust predictions and fast training, while GBR excels in capturing complex relationships and delivering precise predictions. The performance metrics highlight the models' ability to predict key greenhouse parameters with high accuracy. Future work should explore the integration of reinforcement learning to make device control more adaptive and dynamic. Additionally, the development of hybrid models that combine machine learning with domain-specific biophysical models could further enhance reliability and effectiveness. Addressing the complex, dynamic, and nonlinear nature of greenhouse environments remains a priority for advancing intelligent agricultural systems.

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